

Senior High School

General Biology II

Quarter 1 – Module 1.1

Genetic Engineering



S L M

SELF-LEARNING MODULE

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Introductory Message

For the facilitator:

Welcome to the General Biology II 12 Self-Learning Module on Genetic Engineering.

This module was collaboratively designed, developed and reviewed by educators both from public and private institutions to assist you, the teacher or facilitator in helping the learners meet the standards set by the K to 12 Curriculum while overcoming their personal, social, and economic constraints in schooling.

This learning resource hopes to engage the learners into guided and independent learning activities at their own pace and time. Furthermore, this also aims to help learners acquire the needed 21st century skills while taking into consideration their needs and circumstances.

In addition to the material in the main text, you will also see this box in the body of the module:



Notes to the Teacher

This contains helpful tips or strategies that will help you in guiding the learners.

As a facilitator, you are expected to orient the learners on how to use this module. You also need to keep track of the learners' progress while allowing them to manage their own learning. Furthermore, you are expected to encourage and assist the learners as they do the tasks included in the module.

For the learner:

Welcome to the General Biology II 12 Self-Learning Module on Genetic Engineering.

This module was designed to provide you with fun and meaningful opportunities for guided and independent learning at your own pace and time. You will be enabled to process the contents of the learning resource while being an active learner.

This module has the following parts:

Activity	: This will bring understanding to what you already know and experience to what you should learn further.
Analysis	: In this phase, you will process and classify what is valid and not for a more in-depth understanding.
Abstraction	: This part leads you in reinforcing what you know and should know more. Exercises are presented for independent practice to solidify your understanding and skills of the topic.
Application	: This stage brings you to a more practical way that you are going to use what you have learned and think new ways on how it can be improved further.

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CONTENT STANDARD

The learners outline the steps in Recombinant DNA.

PERFORMANCE STANDARD

The learners shall be able to explain how genes may be modified and/or inserted in host cells organisms.

LEARNING COMPETENCY

- The learners should be able to outline the steps involved in genetic engineering (**STEM_BIO11/12-III a-b-6**)

LEARNING OBJECTIVES:

At the end of the module, you should be able to:

1. describe general techniques used by genetic engineers to modify DNA;
2. value the importance by listing several present day applications of genetic engineering; and
3. analyze the benefits and drawbacks of manipulating an organism's DNA.

INTRODUCTION

This self-guided module is focused on understudies in the 'new normal' instructive setting as endorsed by the Department of Education. It tends to a wide scope of learning targets intended to advance students' information and sensational learning involvement with General Biology II. With its credible and genuine exercises, you get skill through picking up bits of knowledge and comprehension of the idea of life science for more profound appreciation.

The module advances autonomous learning of the objective subjects, ideas, and abilities that will build up your 21st century genuine based aptitudes. This module gives you significant assignments to build up your abilities for scholarly achievement and the universe of work. It is secured on the overall standards, objectives, and goals of the K to 12 Basic Education program for Grade 12 that will empower you to become self-realizing, profitable and viable member of the general public and the world on the loose.

PRE-TEST

Directions: Read each item carefully and identify what is asked or described. Encircle the letter of the correct answer.

1. In genetic engineering, what is transferred from one organism to another to produce a genetically modified organism?
 - A. Deoxyribonucleic Acid (DNA)
 - B. Enzymes
 - C. Genes
 - D. Proteins
2. What is the name of the structure found in bacterial cells into which genes may be inserted?
 - A. Enzymes
 - B. Plasmid
 - C. Protein
 - D. Ribosome
3. In genetic engineering what is used to cut the required gene out of the chromosome?
 - A. DNA
 - B. Enzymes
 - C. Genes
 - D. Proteins
4. Which of the following is NOT an argument in favor of Genetically Modified Organisms?
 - A. Controlled production of insulin
 - B. Disease resistant crops
 - C. Food with extra nutrients
 - D. Reduced biodiversity
5. The process of making changes in the DNA code of a living organism is called _____.
 - A. Genetic Engineering
 - B. Hybridization
 - C. Inbreeding
 - D. Selective Breeding

6. One advantage of genetic engineering for agriculture is _____
- A. Cloning plants
 - B. Curing genetic diseases
 - C. Increasing yield
 - D. Mass production of plants
7. Researchers have been working on cloning animals such as sheep. Which result is most likely a negative result of cloning?
- A. Low rate of genetic diseases
 - B. Delayed aging
 - C. Development of extra organs
 - D. Premature death
8. Which of the following is not associated with genetic engineering?
- A. Cloning
 - B. Expression
 - C. Transformation
 - D. Translation
9. Which of the following is the salient feature of genetically modified organisms?
- A. Are always harmful
 - B. Are always micro-organisms
 - C. Arise naturally
 - D. May be beneficial
10. What is recombinant DNA?
- A. DNA that has been sequenced
 - B. DNA that causes genetic disorders
 - C. DNA which has been changed over generations by natural selection
 - D. Molecules of DNA from two different species that are inserted into a host organism to produce new genetic combinations
11. Which of the following is NOT an issue associated with genetic engineering?
- A. Introduction of toxins into the human body
 - B. Increased exposure to vitamins and minerals
 - C. The development of new allergies
 - D. Pests developing genetic resistance
12. In agriculture, how has genetic engineering helped to manage unwanted insects?
- A. By developing new insecticides
 - B. By inserting the genes for bacteria into the crop
 - C. By using genetically modified chemicals
 - D. By deleting gene sequences in the crop

13. Which of the following statements best describes genetic engineering?
- A. It is an alteration in the genetic makeup of an organism is by inserting or deleting specific pieces of DNA
 - B. It is an alteration in the genetic makeup of an organism is by deleting or changing specific pieces of DNA
 - C. It is an alteration in the genetic makeup of an organism is by inserting, deleting, or changing specific pieces of DNA
 - D. It involves an alteration in the genetic make-up of an organism through inserting, deleting or changing specifies pieces of DNA
14. What is the ultimate source of genetic variability?
- A. Evolution
 - B. Genetic drift
 - C. Haplotype
 - D. Mutations
15. Which of the following steps is NOT essential in producing recombinant DNA?
- A. Read the sequences of bases in of DNA
 - B. Cut out a piece of DNA from a DNA molecule
 - C. Use a restriction enzyme to cut DNA and from sticky ends
 - D. Insert a piece of DNA from one organism into the DNA of another organism

“Congratulations you made it!

If you are confident that all your answers are correct, very good! This shows that you already know much about the topics in this module. You may still study the module to review what you already know. Who knows, you might learn a few more new things as well.

If you think you got a low score, don't feel bad. This means that this module is for you. It will help you to understand some important concepts that you can apply in your daily life. If you study this module carefully, you will learn the answers to all the items in the test and a lot more!

Are you ready? Let's go!”

PRESENTATION OF THE MODULE

Genetic engineers have created hereditary recombination procedures to control quality groupings in plants, creatures and different living beings to communicate explicit attributes. Applications for hereditary designing are expanding as engineers and researchers cooperate to distinguish the areas and elements of explicit qualities in the DNA arrangement of different life forms. When every quality is grouped, engineers create approaches to adjust them to make creatures that give advantages, for example, cows that produce bigger volumes of meat, fuel-and plastics-producing microorganisms, and nuisance safe yields.

In this module, you will figure out how specialists apply their comprehension of DNA to control explicit qualities to create wanted characteristics, and how engineers have utilized this training to address current issues confronting humankind. You will realize additionally what hereditary designing methods and instances of its applications, just as good and moral issues identified with its execution. Additionally, you will round out a stream outline to list the strategies to adjust qualities to make GMOs and model utilizations of microorganisms, plant and creature GMOs.

ACTIVITY

“SEARCH THE WORD”

Directions: Locate the given words in the grid and underline, running in one of eight possible directions horizontally, vertically, or diagonally. Have fun searching!

P	Q	A	N	D	U	W	E	F	H	A	O	K	E	B	M	X	Q	L	E	T	D	A	R
I	V	Y	F	B	I	B	Q	V	E	B	F	T	W	T	U	J	X	Z	P	R	R	A	O
X	W	T	F	T	B	N	Y	V	A	K	O	R	R	K	T	C	P	C	E	V	A	C	P
A	S	G	I	U	B	G	U	X	X	G	H	W	G	X	A	L	O	O	H	B	H	C	Y
F	K	E	C	X	P	X	B	D	Z	O	P	F	O	E	T	O	X	S	S	Z	Y	Z	J
D	Z	N	H	K	L	S	N	D	J	L	X	Y	B	M	I	N	T	I	I	M	J	C	S
W	D	E	N	S	M	S	I	N	A	G	R	O	I	O	O	I	I	I	F	Q	F	E	A
V	X	S	D	N	I	E	Y	F	X	L	V	A	E	N	N	N	C	C	O	M	H	X	H
D	G	P	H	X	S	I	M	L	V	Q	V	K	A	E	G	G	X	I	L	R	K	O	P
N	Y	L	S	K	Y	P	R	D	H	G	N	D	U	G	L	C	J	T	G	M	B	D	Q
A	H	I	Z	G	K	G	J	W	S	C	A	S	T	I	A	R	T	B	O	E	A	U	S
M	Z	C	Y	B	U	G	N	O	I	T	A	C	I	F	I	D	O	M	P	T	C	N	V
U	Z	I	D	W	K	Z	B	I	W	Z	Q	M	D	A	E	L	Z	Y	Z	A	F	T	Z
H	Y	N	Q	G	F	V	G	W	K	Q	F	R	C	P	K	U	A	V	Z	L	V	X	C
R	O	G	L	Q	A	V	Z	Y	U	X	S	N	B	D	T	R	V	C	Y	U	T	U	B
E	T	D	L	T	F	U	E	I	K	D	I	M	I	D	W	O	Z	U	Q	P	R	O	V
P	N	O	I	T	A	T	N	E	M	I	R	E	P	X	E	G	N	L	W	I	A	I	T
U	T	V	O	E	T	V	R	A	X	L	J	Z	K	X	Q	H	L	C	I	N	N	F	I
S	T	A	S	Z	G	E	I	Y	J	Y	H	U	H	L	L	L	L	O	O	A	S	V	P
X	G	G	I	E	M	C	L	D	W	F	U	L	G	V	H	W	J	L	G	M	G	U	T
J	S	D	W	G	Z	T	I	O	I	M	F	D	C	E	F	I	O	U	I	N	E	Z	T
M	V	Y	F	E	H	O	E	P	O	H	X	G	S	E	N	E	G	W	B	A	N	U	E
H	S	R	R	W	C	R	W	P	T	N	W	H	G	E	N	O	M	E	S	N	I	Y	W
W	K	V	T	C	R	B	W	T	E	V	B	V	O	H	N	V	F	D	E	Z	C	P	L

transgenic	vector	traits	organisms
genesplicing	cloning	modification	DNA
experimentation	genes	genome	genomes
glofish	manipulate	mutation	superhuman

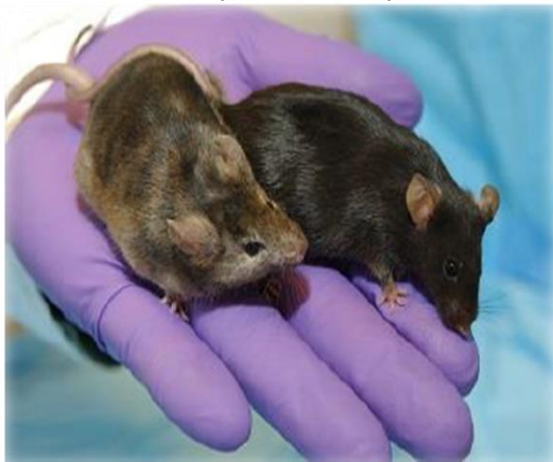
“I know you can do it. Now it’s time for you to EXPLORE the mystery behind those words. You’re doing great!”

ANALYSIS

“GUESS WHAT?”

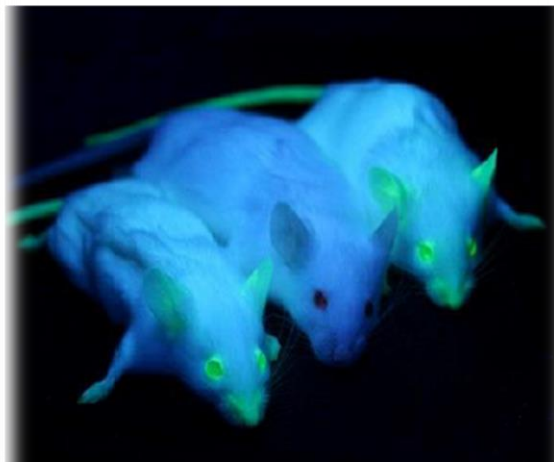
Directions: Below are pictures of laboratory rats which are bred and kept for scientific research. Based on your observation, what is the difference between Picture A and Picture B? Write your observations in the space provided and state possible reasons as to the connections/relevance of the words you had sought in the previous activity.

Picture A
Normal Laboratory Rats
(Black Color)



Source: https://www.teachengineering.org/content/uoh_/lessons/uoh_genetic/uoh_genetic_lesson01_presentation_v2_tedl_dwc.pdf

Picture B
Albino Laboratory Rats



Source: https://www.teachengineering.org/content/uoh_/lessons/uoh_genetic/uoh_genetic_lesson01_presentation_v2_tedl_dwc.pdf

Good job! I told you, you can do better! Now, it is time for you to know it all. Read on.

ABSTRACTION

Deoxyribonucleic acid (DNA) may be a large biomolecule that contains the entire genetic information for an organism. Every cell of living organism and lots of viruses, contains DNA. The fundamental building block of a DNA molecule is termed a nucleotide, and one strand of DNA may contain billions of nucleotides. Although each DNA molecule contains many of those building blocks, only four unique nucleotides are designed to create the whole DNA sequence; these are written as A, G, C and T. These four nucleotides will be arranged in sequences to "spell" the genetic instructions to make all of the various proteins organisms must live.

Because DNA contains instructions for an organism to form several different proteins, it's useful to define another sub-unit of DNA called **genes**. Each gene may be a small segment of DNA that contains a group of instructions for an organism to form one protein; one organism may have thousands of various genes. Together, the complete set of genes for an organism is named its genome. To use another analogy, think about the genome as a whole cookbook for an organism, and every gene is a private recipe therein cookbook. When one recipe is followed, the result's a particular protein.

Why are proteins important?

Proteins perform all of the work in organisms. Some functions of proteins include:

- Serving as catalysts for reactions
- Performing cell signaling
- Transporting molecules across membranes
- Creating structures

When a protein is made by its gene, it's said that the gene is "expressed," or used. Most gene expressions don't produce results visible to the unaided eye. However some genes, like those who code for proteins answerable for pigment, do have visual expression. The expression of a gene in an observable manner is named a **phenotypic trait**; one example is an organism's hair color. In fact, everything you'll see in an organism could be a result of proteins or protein actions.

How is DNA employed in gene-splicing?

The process of using plants or animals with specific traits to breed offspring's with those traits is thought as selective breeding, something that has been practiced since the first times. **Hybridization**, or the method of crossing plants or animals with different variations of the identical trait within which the resulting offspring is formed to

possess best traits of the parents, has been around for several years yet. However, these processes require an extended period before results can be seen and frequently succeed only within a detailed family of species. Within the early 1970s, biologists have realized that they'll manipulate and deliberately recombine the DNA molecules from different species in laboratory. The technique, **genetic engineering**, allow scientist to provide more complex solutions to problems within the society. The term "biotechnology" came from the words biology and technology. it had been coined within the 1970s when the primary genetically engineered bacteria were reported. Since then, biotechnology has often been related to gene-engineering, specifically within the development of genetically engineered microorganisms.

By definition, genetic engineering is that the direct altering of an organism's genome. This can be achieved through manipulation of the DNA. Doing this can be possible because DNA is sort of a universal language; all DNA for all organisms is formed of the identical nucleotide building blocks. Thus, it's possible for genes from one organism to be read by another organism. Now imagine that everyone's cookbooks are written within the same language; thus, any recipe are often inserted and employed in the other cookbook.

In practice, since DNA contains the genes to make certain proteins, by changing the DNA sequence, engineers are ready to provide a brand new gene for a cell/organism to form a distinct protein. The new instructions may supplement the old instructions such an additional trait is exhibited, or they'll completely replace the old instructions specified a trait is modified.

Genetic Engineering Technique

The process for genetic engineering begins the identical for any organism being modified.

1. Choose an organism that contains a desirable gene.
2. Extract the whole DNA from the organism.
3. Remove this gene from the remainder of the DNA. A way to try this can be by employing a restriction enzyme. These enzymes seek for specific nucleotide sequences where they'll "cut" the DNA by breaking the bonds at this location.
4. Insert the new gene to an organism's DNA.
5. When modifying bacteria, the foremost common method for this final step is to feature the isolated gene to a plasmid, a circular piece of DNA employed by bacteria. This is often done by "cutting" the plasmid with the identical restricted enzyme that was needed to remove the gene from the initial DNA. The new gene can now be inserted into this opening within the plasmid and also the DNA are often bonded back together using another enzyme called ligase. During this case, the recombinant plasmid is additionally named as a bacterial artificial chromosome (BAC).

Once the recombinant DNA has been built, it is passed to the organism to be modified. If modifying bacteria, this process is kind of simple. The plasmid may be easily inserted into the bacteria where the bacteria treat it as their own DNA. For plant modification, certain bacteria like *Agrobacterium* species is also used because these bacteria permit their plasmids to be passed to the plant's DNA.

Applications and Economics

The quantity of utilizations for hereditary building are expanding as increasingly more is found out about the genomes of various living beings. Some of fascinating or outstanding application zones are depicted beneath.

What number of the present harvests are hereditarily changed? Starting at 2010, in the U.S., 86% of corn created was hereditarily adjusted. Bt-corn is a typical GMO that joins a quality from the Bt microbes with corn DNA to deliver a harvest that is bug safe. The microbes' quality utilized contains a formula for a protein that is harmful when devoured by bugs, however safe when devoured by people.

Various different qualities can be joined with yields to create alluring properties, for example,

- Herbicide-, dry season , freeze-or disease resistance
- Higher yield
- Faster development
- Improved sustenance
- Longer time span of usability

The formation of hereditarily altered harvests gives numerous motivators to farmers and organizations. At the point when farmers can plant a harvest that has a better return for every section of land, they can essentially expand production, and in this manner deals, with insignificant expense. Sickness, bug and different protections lessen crop misfortune, which likewise assists with expanding benefits. Other than farmers, different promoters from adjusted yields incorporate seed, agrochemical and horticulture hardware organizations just as merchants and colleges that are associated with GMO research.

Because of their basic structures, the most regularly adjusted living beings are microorganisms. The main adjusted microbes were made in 1973. Microorganisms can be altered to create alluring proteins that can be reaped and utilized. One model is insulin or insect silk, which is hard to accumulate normally. Different adjustments to microorganisms incorporate creation changes to the cellular respiration cycle to modify the results; ordinarily CO₂ is delivered, anyway builds have made alterations so

hydrocarbon side-effects, for example, diesel and polyethylene (a fuel and a plastic) are delivered.

Morals/Ethics

The primary explanation hereditarily adjusted/genetically modified living beings are not all the more generally utilized is because of moral concerns. Almost 50 nations around the globe, including Australia, Japan and the entirety of the nations in the European Union, have established noteworthy limitations or full prohibitions on the creation and offer of hereditarily changed creature food items, and 64 nations have GMO naming prerequisites. A few issues to consider when concluding whether to make or potentially use GMOs include:

Wellbeing: This for the most part emerges on account of GMO nourishments. Are the nourishments alright for human utilization? Is GMO feed solid for creatures? Numerous adversaries of GMO nourishments state insufficient autonomous testing is done before the food is affirmed available to be purchased to customers. When all is said in done, research has indicated that GMO nourishments are alright for people. Another wellbeing thought is the soundness of farmers and their families, animals and networks who are put in danger with presentation to synthetic compounds utilized pair with GMO seeds.

Natural Impact: Consider that hereditary designers can make trees that become quicker than their unmodified partners. This appears to be a lot for the wood business, yet may some unintended outcomes result? Being outside and developed in enormous amounts, the adjusted trees may cross-fertilize with unmodified trees to shape half and halves outside of assigned developing territories. This consequently could make trees that could upset the environment. For instance, they could overpopulate the zone or develop so huge that they cover other vegetation. This equivalent situation has unintended and unwanted results when the dust from GMO crops floats into non-GMO fields.

People: Should people be hereditarily built? Doing so could have clinical applications that lessen or forestall hereditary issues, for example, Down's syndrome. In any case, the greater inquiry is the place should designing people stop? Should guardians be permitted to choose their kids' eye hues, statures or even sexes before birth?

Additional Information:

1. Classical rearing practices center around the mating of life forms with attractive characteristics.

2. Genetic Engineering includes the utilization of atomic strategies to alter the attributes of an objective living being. The adjustment of attributes may include:
 - i. presentation of new attributes into a creature
 - ii. upgrade of a current attribute by expanding the statement of the wanted quality
 - iii. upgrade of a current attribute by upsetting the restraint of the wanted qualities' demeanor.

3. An overall layout of recombinant DNA might be given as follows:
 - i. cutting or cleavage of DNA by restriction enzymes (REs)
 - ii. choice of a suitable vector or vehicle which would proliferate the recombinant DNA (e.g. roundabout plasmid in microbes with an unfamiliar quality of intrigue)
 - iii. ligation (combine) of the quality of intrigue (e.g. from creature) with the vector (cut bacterial plasmid)
 - iv. move of the recombinant plasmid into a host cell (that would convey out replication to make enormous duplicates of the recombined plasmid)
 - v. determination cycle to screen which cells really contain the quality of intrigue
 - vi. sequencing of the quality to discover the essential structure of the Protein

4. Manners by which these plasmids might be brought into host creatures.

Biolistics. In this strategy, a "gene gun" is utilized to fire DNA-covered pellets on plant tissues. Cells that endure the siege, and can take up the articulation plasmid covered pellets and gain the capacity to communicate the planned protein.

Plasmid insertion by Heat Shock Treatment. Heat Shock Treatment is a cycle used to move plasmid DNA into microorganisms. The objective cells are pre-treated before the technique to expand the pore sizes of their plasma layers. This pretreatment (as a rule with CaCl_2) is said to make the cells "equipped" for tolerating the plasmid DNA. After the cells are made capable, they are brooded with the ideal plasmid at about 4°C for about 30min. The plasmids concentrate close to the cells during this time. A while later, a "Heat Shock" is done on the plasmid-cell arrangement by hatching it at 42°C for brief at that point back to 4°C for 2 minutes. The quick ascent and drop of temperature is accepted to increment and abatement the pore sizes in the layer. The plasmid DNA close to the film surface are taken into the cells by this cycle. The cells that took up the plasmids obtain new attributes and are supposed to be "changed".

Electroporation. This procedure follows a comparable technique as Heat Shock Treatment, yet, the extension of the layer pores is done through an electric "shock". This strategy is normally utilized for addition of qualities into mammalian cells.

5. A few strategies to screen recombinant cells are as per the following:

Determination of plasmid DNA containing cells

A determination marker inside the embedded plasmid DNA grouping permits the choice of "transformants". Typically, an anti-microbial obstruction quality (for example AMP ampicillin resistance gene) is remembered for the plasmid DNA. This permits just "changed" cells to make due within the sight of the anti-microbial (for example ampicillin). Plating the plasmid-cell arrangement on anti-toxin containing media will choose for these "transformants" and just permit plasmid-containing cells to develop and engender into states.

Choice of changed cells with the ideal quality

Certain embedded qualities inside the plasmids give obvious evidence of their essence. These incorporate the anti-microbial resistance genes that take into consideration the determination of the changed cells inside the arrangement.

Some embedded qualities additionally produce shaded (for example chromogenic proteins) or fluorescent items (for example GFP) that name the states/cells with the embedded quality. Now and again, the area of the cloning site inside the plasmid is in a quality (for example β -galactosidase, lacZ) that creates a (blue) colored item within the sight of a substrate (for example isopropyl β -D-1 thiogalactopyranoside, or IPTG). Cells changed with these "unfilled" plasmids will turn blue within the sight of IPTG. Inclusion of a quality in the cloning site disturbs the succession of the β -galactosidase quality and forestalls the age of the hued item within the sight of the substrate. Cells changed with the upset β -galactosidase quality will stay "white" within the sight of IPTG. This "blue-white screening" convention is hence ready to screen for cells that were changed with the ideal quality in the cloning site.

PCR identification of plasmid DNA

Then again, the presence of the ideal quality in the embedded plasmids might be affirmed utilizing PCR enhancement. PCR responses explicit for the ideal quality might be finished utilizing DNA from cells. Intensification of the normal item would affirm the presence of the quality inside the examples. PCR responses explicit for plasmid arrangements will likewise affirm/recognize the kind of plasmid utilized for the change.

Genetically Modified Organisms (GMOs)

With the capacity to embed quality arrangements, comes the chance of giving new characteristics to these objective creatures. This has permitted the advancement of GMOs. A portion of these hereditary changes guarantee higher item yield for their objectives. These incorporate the Flavr-Savr Tomato and Bt-Corn.

The Flavr-Savr ("Flavor Savor") tomato was the primary hereditarily altered creature that was authorized for human utilization. The characteristic changed in this tomato is its aging cycle. A quality for a compound that causes the debasement of gelatin in the cell dividers (for example polygalacturonase) typically mellow the organic product as it matures. In Flavr Savr tomatoes, an inhibitor (for example antisense RNA) disturbs the statement of this quality, consequently postponing the conditioning of the products of the soil the time it might be kept away and moved to business sectors.

Bt-Corn was created to consolidate the creation of a poison (for example Bt-endotoxin) from *Bacillus thuringensis* in corn plants. This poison brings about the passing of nuisances that feed on these plants like the corn drill hatchlings. The poison has been demonstrated to be specific for *Lepidoptera* hatchlings and is non-harmful to people, well evolved creatures, fish and winged animals. The particular harmfulness of the poison permits its utilization in food crops. The presentation of the poison is accepted to expand crop creation because of diminished misfortunes from bug invasion. A similar innovation has been applied in the Philippines for the advancement of Bt-Eggplant.

Notwithstanding the proposed advantages of GMOs, a few people have raised their interests with respect to the utilization of these adjusted nourishments. While the greater part of the items are tried for security, concerns are raised for the chance of not having the option to identify perils that are available, however are as of now imperceptible by the present current innovation.

Due to these issues, makers are encouraged to give names that inform purchasers of GMO presence in their items. While GMOs are accepted to be sheltered when authorized by the food administrative organizations, it is accepted that the customers must be given enough data to settle on their own decisions with respect to their utilization.

APPLICATION

Directions: Answer the following questions based on your understanding.

1. What is your opinion about Genetic Engineering? List down (at least five, ten, or more) its pros and cons.

That was excellent! Keep it up!

REFLECT

You did it well! Congratulations on finishing the first supplementary learning module in General Biology II! You have just had an amazing learning journey and for sure, you will also do the same in the succeeding modules.

For the last time, share your final insights by completing the following sentence prompts. It was such a wonderful learning experience with you.

3	Things I learned today...
2	Things I found interesting...
1	1 Question I still have...

LEARN MORE!

Since learning is considered as a continuous process, I have a task for you; please have time to work on it. Goodluck!

- Watch and make a short reflection about the video
“Genetic Engineering Will Change Everything Forever – CRISPR”
<https://www.youtube.com/watch?v=jAhjPd4uNFY>.

REFERENCES

Ramos, A., (2017). Exploring life through science series: General Biology 2. Phoenix Publishing House, pp. 238-235.

Teaching Guide for Senior High School-General Biology 2 (2016) Commission on Higher Education, pp. 41-46

Online Resources

https://www.teachengineering.org/lessons/view/uoh_genetic_lesson01

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